

Unified Money-Trail Risk Detection System

Bank Transaction Pattern Analysis + PS-12 Cash Withdrawal / ATM Risk Prediction

Hackathon Project — Full Technical & Concept Report

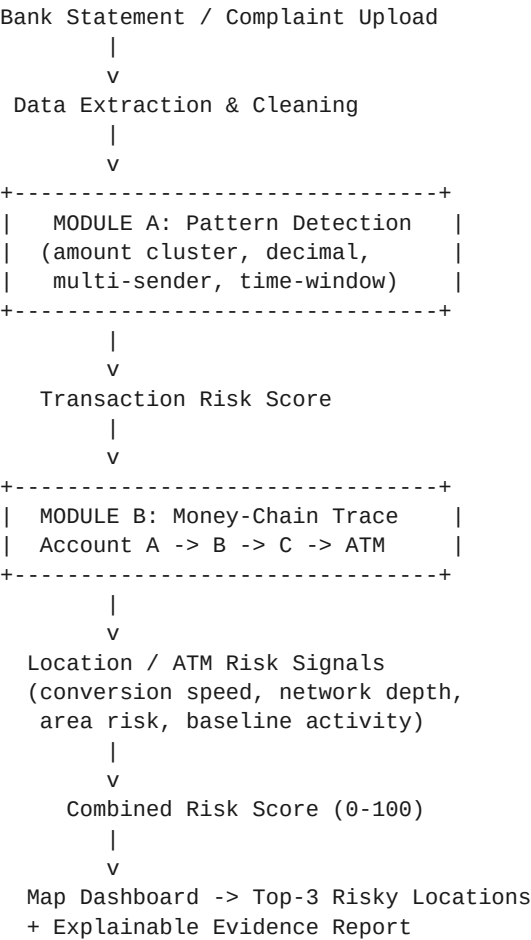
"Hamara system bank-statement patterns se suspicious money movement detect karta hai, uske aage money-chain ko trace karta hai, aur end me high-risk ATM/Micro-ATM withdrawal locations ko map par highlight karta hai."

1. Combined Project Vision

Do alag modules ko ek hi pipeline me jodna hai:

- Module A — Pattern Detection (Bank Statement Level):** Uploaded statement me suspicious behavioural patterns dhundna (amount clustering, multi-sender concentration, time-window anomalies, decimal patterns).
- Module B — Money Trail → ATM Risk Prediction (PS-12):** Complaint ke baad paisa kis route se, kitni jaldi withdraw hua, aur kaunsi ATM/Micro-ATM location par risk zyada hai — ye predict karna.

2. Combined End-to-End Flow



3. Signal Inventory (Dono Modules Se Combined)

Layer	Signal	Simple Meaning
Module A	Amount Cluster Detection	Same range ke amounts multiple senders se aana
Module A	Decimal / Non-Round Pattern	■99.50 jaise odd amounts baar-baar
Module A	Multi-Sender Concentration	Ek hi receiver ko many senders credit karte hain
Module A	Time-Window Clustering	Short window me repeated multi-sender activity
Module A	Money-Chain (fan-in/out/cycle)	Graph me suspicious flow structure
Module B	Conversion Speed	Complaint se withdrawal tak kitna time laga (kam = risky)
Module B	Network Depth	Paisa kitne accounts se hokar gaya (zyada hops = risky)

Layer	Signal	Simple Meaning
Module B	Area Risk	ATM/area ka past fraud-linked withdrawal history
Module B	Baseline Activity	ATM ka normal daily volume (busy ATM galat flag na ho)

4. Two-Stage Combined Scoring

Stage 1 — Transaction Pattern Score (Module A)

$\text{score_A} = (\text{decimal_pattern} \times 10) + (\text{repeated_txn} \times 15) + (\text{splitting} \times 20)$
 $+ (\text{multi_sender} \times 20) + (\text{rapid_transfer} \times 15)$
 $+ (\text{unusual_merchant} \times 15) + (\text{circular_flow} \times 25)$

Thresholds: 0–20 Low, 21–40 Medium, 41–60 High, 61+ Critical

Stage 2 — Location/ATM Risk Score (Module B)

$\text{score_B} = (\text{network_risk} \times 0.4) + (\text{area_risk} \times 0.3)$
 $+ (\text{time_risk} \times 0.2) - (\text{baseline} \times 0.1)$

Thresholds: 0–40 Low, 41–65 Medium, 66–85 High, 86+ Critical

(Weights illustrative/starting hain — validated nahi, presentation me clearly bolna)

Combined Final Risk

$\text{final_score} = \text{weighted_average}(\text{score_A}, \text{score_B})$
 -> agar dono modules high aa rahe hain (statement pattern + ATM location)
 to combined confidence sabse strong hoga

5. Pages / Screens (Map Module Ke Saath)

Page	Purpose
Landing Page	Combined project intro + disclaimer
Upload Page	Bank statement / complaint data upload
Processing Page	Extract → Clean → Pattern Detect → Chain Trace → Location Risk → Score
Transaction Dashboard	Module A output: clusters, evidence table, graph view
Map Dashboard (new)	Module B output: H3 hexagon map, Top-3 risky locations, click-to-see reason
Combined Report Page	Both module scores + explainable reasoning
About / Methodology	Formula transparency + disclaimer

Map Dashboard Details (from PS-12)

- ATM locations → H3 hexagons
- Color-coded by risk: Critical / High / Medium
- Hover/click → shows score + reason (e.g. "Risk 89 – high network fan-out + fast conversion")
- Suggested tools: h3-py, Streamlit or Folium+Flask for map layer; React frontend via Leaflet/Mapbox

6. Data Model (Merged)

Table	Fields
transactions	id, date, time, amount, sender, receiver(masked), type
accounts	account_id(masked), first_seen, last_seen, total_in, total_out
complaints	complaint_id, victim_account, amount, timestamp
money_chain	chain_id, path(account sequence), hop_count, total_time
atm_locations	atm_id, lat, lng, h3_index, baseline_volume
signals	id, ref_type(txn/account/atm), signal_type, weight, detected_at
reports	id, ref_id, score_A, score_B, final_score, risk_level, generated_at

7. Folder Structure (Updated)

```
project-root/
|-- frontend/
|   |-- pages/
|   |   |-- Landing.jsx
|   |   |-- Upload.jsx
|   |   |-- Processing.jsx
|   |   |-- TransactionDashboard.jsx
|   |   |-- MapDashboard.jsx      <- new
|   |   |-- CombinedReport.jsx   <- new
|   |   |-- About.jsx
|   |-- components/
|   |   |-- RiskBadge.jsx
|   |   |-- ClusterChart.jsx
|   |   |-- GraphViz.jsx
|   |   |-- EvidenceTable.jsx
|   |   |-- MapView.jsx          <- new (H3/Leaflet)
|-- backend/
|   |-- routes/
|   |   |-- upload.js
|   |   |-- analyze.js
|   |   |-- moneyChain.js        <- new
|   |   |-- atmRisk.js          <- new
|   |   |-- report.js
|   |-- services/
|   |   |-- detectClusters.js
|   |   |-- detectDecimalPattern.js
|   |   |-- detectMultiSender.js
|   |   |-- timeWindowAnalysis.js
|   |   |-- graphBuilder.js
|   |   |-- conversionSpeed.js   <- new
|   |   |-- networkDepth.js      <- new
|   |   |-- areaRisk.js          <- new
|   |   |-- riskScoring.js       (combined scoring)
|-- README.md
```

8. Security & Data Notes (Combined)

- OTP/UPI PIN/password kabhi collect nahi karna
- Aadhaar/phone jaise sensitive real data use nahi karna — fake/hashed identifiers
- Sensitive identifiers masked display me hi
- MVP me synthetic/fake data use karna (20–30 fake fraud chains + normal records) — real police/bank data nahi
- Accuracy % sirf tab claim karna jab synthetic data par actually calculate kiya ho

9. 24-Hour Hackathon Plan

Time	Task
0–2 hr	Roles divide + fake/synthetic data (transactions + complaints + ATM)
2–6 hr	Module A signals (cluster, decimal, multi-sender) + Module B features (conversion speed, network depth, area r
6–11 hr	Map/Dashboard + Transaction Dashboard build
11–15 hr	Integration: Module A → Module B → combined scoring, end-to-end testing

Time	Task
15–19 hr	UI polish + explanation text + bug fixing
19–21 hr	Deck + README
21–23 hr	Demo video
23–24 hr	Final links/files check + submission

10. Important Disclaimer

Ye ek hackathon MVP / decision-support prototype hai — proven real-world fraud detector nahi. Suspicious pattern illegal activity ka proof nahi hota. Final determination sirf appropriate evidence aur authorized investigation se hoga. Real deployment ke liye thresholds aur weights validated data se calibrate karne honge.